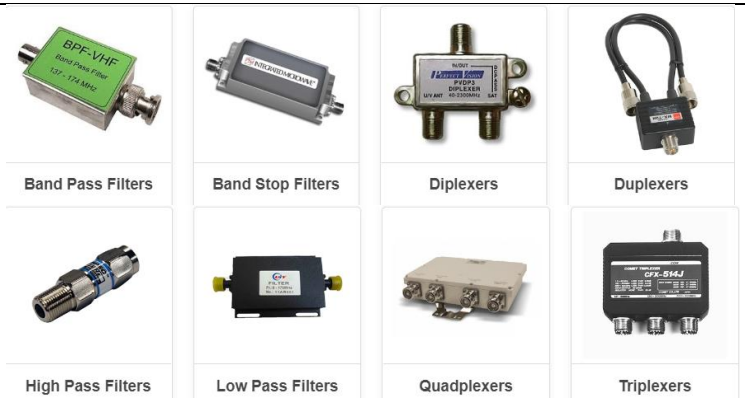
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Kelas : 3NK

RF & MICROWAVE DEVICES

No	Item	Gambar	Spesifikasi	Aplikasi/Kegunaan
1.	Filters		**Filter Type:** There are a number of different types of RF filters - Band Pass Filters, Low Pass Filters, Band Stop Filters, High Pass filters etc. Each type operates in a different way. **Technology:** Based on the required application and size of the wireless system there are a number of filter types - Notch Filters, SAW Filters, Cavity Filters, Waveguide Filters etc. Each one has different properties and different form factors. **Passband Frequency (MHz):** This is the frequency range where signals can pass through with minimal attenuation. **Stopband Frequency (MHz):** This is the frequency range where the signals are attenuated. The higher the attenuation the better. This is also called isolation. **Insertion Loss (dB):** It is the loss that occurs while a signal is traveling	RF and Microwave Filters are used to filter out unwanted signals from entering a system. With the increase of wireless standards in the existing frequency bands, filters now play an extremely important role and are required to minimize interference. They are designed to operate at specific frequencies and allow/attenuator RF signals at different frequencies. RF Filters have two types of frequency bands - passband and stopband. Signals which lie in the passband can pass through with minimal attenuation while signals which lie in the stopband experience heavy attenuation.

			through the passband frequency range. The lower the insertion loss the better the filter performance. **Stopband Attenuation (dB):** It is the attenuation experienced by signals which lie in the stopband of a given filter. The magnitude of attenuation faced by signals can vary according to their frequency.	
2.	Cables		**Cables:** The cable that connects the two connectors can be selected from here. The length of the rf cable assembly can also be selected using the 'Length' filter. **Connector Type:** Connector 1 and Connector 2 are the connectors at either end of the cable. **Connector Gender:** The gender of both the Connectors can be selected under the filters Connector 1 Gender and Connector 2 Gender. **Connector Geometry:** The Geometry of the connectors in the cable assemblies can be selected using the Connector 1 Geometry and Connector 2 Geometry filters.	RF Cable Assemblies are high frequency and mmWave cables with coaxial connectors on either end. Cable Assemblies are selected based on the connector interfaces and the length of the cable. There are cases where the same coaxial connector is required on both ends, and there are cases where different coaxial connectors are required on either end.

			Frequency: This is the frequency up to which the RF Cable Assembly can operate. This is based on the Connectors and Cable used. View the RF Connector Frequency Chart on everything RF. • Features Designed for co- location of Link 16, TTNT, and IFF, Rugged Compact Design for Harsh Environments • Frequency 962 to 1008 MHz / 1053 to 1065 MHz / 1113 to 1215 MHz	

			• Insertion Loss 0.75 dB • Rejection -60 to -30 dBc • Power 500 W • VSWR 1.7:1 • Package Type Module with Connectors	
3.	Fixtures		• Frequency 0.1 to 15 GHz • Fixture Area 950 mil	Berfungsi meningkatkan kualitas produk ,menghasilkan suatu produk industri yang presisi untuk mencapai kualitas produk yang diinginkan, sekaligus dapat mempercepat dan mempermudah

			• Power Handling 25 W • Type Load Pull • Cooling Type Air, Water • Impedance 50 Ohms • Connectors 3.5 mm - Low Insertion Loss for High VSWR Tuning - Multiple Connector Configurations - 50Ω and Transformers (Klopfenstein, Quarter-wave) Available - Heatsinks and Fans Available - Cooling (Water cooling, Forced air cooling) Available - Integrated Biasing Available	proses pemasangan komponen.

4.	Pads		• Type 50 to 75 Ohms • Frequency 0.009 MHz to 3 GHz • 50 Ohm Connector SMA - Male • Connector SMA, SMA - Male, F Type, F Type - Female • 75 Ohm Connector • F Type - Female • Dimension 64.01 x 20.57 x 20.57 mm • Package Type Connectorized • Attenuation 5.5 to 6.3 dB • VSWR 1.07:1, 1.10:1 • Note Return Loss : 23 to 30 dB, RoHS : Yes	

5.	Mixers		**Frequency:** There are three frequencies that are of importance when selecting a mixer – RF Frequency – This is the high frequency which is usually received by the receiver or transmitted out from a transmitter. LO Frequency – This is the frequency of the local oscillator (LO), it is used to up-convert the Intermedia Frequency (IF + LO) or to down-convert the RF frequency (RF - LO). The Intermediate Frequency (IF) is the low frequency required for processing the signal in the receive chain. **Conversion Loss/Gain:** This is the loss or gain in the amplitude of the signal after frequency conversion. **Isolation:** This is an important parameter, RF/LO Isolation – is the measure of signal leakage from the local oscillator to the RF path and the LO/IF Isolation is the signal leakage from the LO to the IF path. The higher the isolation the better the mixer performance. Type: Double Balanced Mixer RF Frequency:	A RF Mixer is a non linear component that is used to up-convert or down-convert the frequency of an input signal. RF Mixers mix the frequency of an input signal (RF) with that of a Local Oscillator (LO) to produce two new signals at the sum RF + LO and difference RF – LO of the original frequencies. RF Mixers are primarily used in receivers to down-convert a high RF frequency to a lower frequency which is typically easier to process. For example a 2.4 GHz signal input to a receiver can be down-converted to 10 MHz to further process. Similarly it is used in transmitters when up-conversion to a particular higher frequency is required.

			20 to 65 GHz IF Frequency: DC to 20 GHz LO Frequency: 20 to 65 GHz Conversion Loss: 11 to 16 dB P1dB: 10 dBm Package Type: Module with Connectors	
6.	Antennas		• Type Dipole • Directionality Omni-Directional • Application Hand-held and bodyworn tactical MIMO / MANET mesh radio systems, Cellular, unmanned vehicles	Sebagai penangkap sinyal yg membawa sebuah informasi dari gelombang elektromagnetik ke medan yg ada di kabel antenna.

			• Polarization Vertical • Standards ISM • Frequency 1.35 to 2.5 GHz • Bandwidth 1150 MHz (1.15 GHz) • Gain 2.4 dBi • Power 100 W • Vertical BW 58 to 110 Degrees • Horizontal BW 360 Degrees	

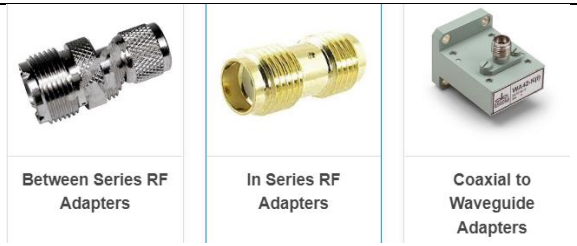
			• Industry Broadcast, Military • VSWR 2.0:1 • Mounting Connector mount • Other features Rugged G10 Fiberglass Radome, Continuous Coverage • Impedance 50 Ohms • Connectors N Type, N Type-Male • Multi Port No • Diameter	

			0.81 in. (20.62 mm) • Length 6.09 in. (154.69 mm) • Weight 3.5 oz (100 gm) • Operating Temperature -40 to 85 Degree C	
7.	Isolators		**Isolation (dB):** The level of Isolation from the output port to the input port. The higher the isolation the better the components are protected. **Insertion Loss (dB):** The signal loss from port 1 to port 2. **Power (W):** This is the power level that the isolator can handle while maintaining its electrical characteristics. The forward power is the level of power that the isolator can handle incident at port 1. The reverse power is the level of power incident at port 2 that the isolator can absorb.	An RF Isolator is a two port device that protects RF components in a system from excessive signal reflection. It is a non-reciprocal device that ensures that all the power is transmitted from port 1 to port 2, while isolating absorbing/isolating any power incident at port 2. An isolator is often placed at the in front of sensitive component sin the Rx/Tx chain to avoid unwanted signals from damaging sensitive components.

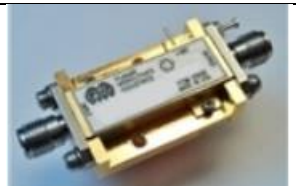
			Configuration: RF Isolators are available in a number of configurations – Drop-In Packages, connectorized modules and surface mount packages. everything RF lists RF Isolators from all the leading manufacturers in the industry. You can use parametric filters to specify the required frequency, isolation and power levels – the search tool will look through multiple manufacturer catalogs to find RF Isolator products that meet your requirement. You can then compare products, download datasheets, request quotations. Quotation request are directed to the manufacturer who will get back to you directly. **M2 Global Technology - 094-039038-025** RF & Microwave Isolator from M2 Global Technology Description:1800 to 2000 MHz, 20 dB, High Power Drop-In Isolator Type: Drop-In Isolator Configuration: Dual Junction Frequency:	

			1.8 to 2 GHz Isolation: 20 dB Average Power Forward: 30 W Average Power Reverse: 30 W Insertion Loss: 0.4 dB	
8.	Couplers		**Types of RF Couplers:** **- Directional Coupler:** This is a four port device – with an input and output port, a port for the coupled signal and a port that is terminated internally. The power that is passing from the input to the output is coupled with a coupled line and passes out through the coupled output port, which is isolated from the main output port. Any reflected power from the main output is coupled to the port that is terminated, resulting in minimal reflections within the device/system. - **Dual Directional Coupler:** This is a four port device where two directional couplers are connected in	Pulsar Microwave manufactures the CS06 and CS08 series single directional couplers in nominal values of 6 and 8 dB, respectively. These directional couplers are used in 50-ohm systems for narrow and wideband applications within 500 MHz to 40 GHz. Standard outlines are fitted with SMA or N type female connectors, but Pulsar Microwave can customize upon request. RF & Microwave couplers are passive devices that are used to sample high

			series, in opposing directions or a directional coupler with a single main line and two secondary lines for coupling. Learn more about Dual Directional Couplers. **- 90° Hybrid Coupler:** 90 degree hybrid couplers split the input signal in to two equal signals with a 90 degree phase differential. **- 180° Hybrid Coupler:** 180 Degree Hybrid couplers split the input signal in to two euqal signals with a 180 degree phase differential. **Frequency (MHz):** This is the frequency range over which the device can operate while providing a constant level of coupling with minimal loss and reflection. **Coupling (dB):** This is the ratio of incident power fed to the main port to the power at the output of the coupled port. **Coupling Variation (dB):** The maximum peak-to-peak variation in coupling expected over the specified frequency range.	frequency signals. It takes one signal as the input and provides two outputs – One being the regular output and the other being the coupled output. Based on the required application, the power level of the coupled signal can be varied when designing the device. Couplers have many applications and are used for sampling signals, signal injection, signal generation, used to measure incident/reflected power to determine VSWR and have a number of other applications.

			Insertion Loss (dB): The reduction in output power due to the power transferred to the coupled line. **Directivity (dB):** The difference between the power level of the coupled and isolated ports. The directivity is a measure of independence of the coupled and isolated ports. **Average Power (W):** The level of power that the device can handle while maintaining its electrical characteristics. **Impedance (Ohms):** The impedance of a coupler must match the circuit/system it is connected to; this will minimize any reflection within the circuit/system. Usually the impedance is 50 Ohms or 75 Ohms.	
9.	Adaptors		RF Coaxial Adapters are are broadly divided in to two types - In Series and Between Series. **<u>In Series RF Adapters</u>:** These adapters have the same connector configuration on either side of the adapter. Some examples of these adapters would be SMA Male to Female, SMA Male to Male, N-Type Male to Female, and soo on. As long	RF Adapters are passive coaxial components that are used to transfer signals from one connector interface to another. They can be used to transition signals from an SMA Connector Interface to an N-Type connector interface of BNC to TNC etc. The Geometry and gender of

			as the connector type is the same the adapters are known as 'In Series Adapters'. The gender on each of the connectors can be different and the geometry of the adapter can be selected to be straight or right angle based on where the adapter will be placed. **Between Series RF Adapters:** These adapters have different connector configurations on either side of the adapter. Some examples of these adapters would be SMA Male to N-Type Female, BNC Female to N-Type Male etc. These RF Adapters have different connector configurations. **MOLEX - 733000223** In Series RF Adapter from MOLEX Description:50 Ohms, SMPM Jack/Jack Bullet Adapter, 8.31mm Length Connector: SMPM Frequency: DC to 65 GHz Adapter - Gender 1: Female Adapter - Gender 2:	each connector interface can vary and be customized to meet the required of a user.

			Female Gender: Female-Female VSWR: 1.25:1, 1.35:1	
10.	Variable Attenuations		Control Voltage: 0 to +8 Volts typical. Operating Power: <= 0 dBm Power Handling: +27 dBm max. Operating Temperature: -25°C to + 80°C Harmonic distortion is affected by input power and frequency. Typical distortion is 50 dBc for input power less than 0.0 dBm. Two tone intermodulation products are typically 54 dBc for input power less than 0.0 dBm. These attenuators are bi-directional. Either J1 or J2 can be used as input.	Our family of voltage controlled attenuators is non-reflective, with ranges of 32 dB of 64 dB. Pin diodes are used and the frequency range is 0.25-16 GHz. They operate with a control voltage of 0 to +8 VDC and handle +27 dBm maximum input power. These attenuators are bi-directional: either SMA port may be used for an input.
11.	Amplifiers		**Type:** There are many types of rf and microwave amplifiers • **RF Power Amplifiers:** Power Amplifiers convert low power high frequency	An RF amplifier is an electronic device that is used to amplify a low power radio frequency signal to one with a larger amplitude. They are used in transmit and receive parts of a wireless system. In the transmit chain they are used to amplify signals before

			signals to high power signals. They are used in the transmit chain to amplify the signal before it is sent out via an antenna. • **Low Noise Amplifiers:** RF Low noise amplifiers are used to amplify high frequency signals while maintaining a good signal to noise ratio. Low noise amplifiers are used at the beginning of the receive chain to amplify a signal with minimal degradation. • **Other Amplifiers:** There are a number of other amplifiers like gain blocks, differential amplifiers, driver amplifiers, limiting amplifiers etc. **Frequency (MHz):** RF amplifiers are usually designed to work at a	they are sent out through the antenna and in the receive chain they are used to amplify weak signals that are captured by the antenna with minimal distortion. RF amplifiers have a wide range of applications in the wireless sector – military, test and measurement, mobile devices and wireless infrastructure.

			particular frequency range. The unit for frequency is Hz (Hertz). **Gain (dB):** The gain of an rf amplifier is the ratio of output to input power or amplitude – It is the measure of amplification of the rf amplifier. **Noise Figure (dB):** Noise is an undesired signal that is generated up to some level in all electronics components and devices. Noise figure is the metric for noise performance in a circuit or system. The lower the noise figure, the higher the signal to noise ratio of a system. **Power Compression (dBm or W):** Power Compression or P1dB is the point at which the input signal is amplified by an amount, which is 1 dB below the small signal gain of the device. It is the point beyond which the gain of the device no longer increases linearly as the input power is raised. **Package Type:** RF amplifiers are available in surface mount packages, as a die / chip, in a module with	

			connectors, as a system and in a rack mount enclosure. **<u>Quantic PMI (Planar Monolithics) - PE2-30-24R526R5-8R0-24-12-SFF</u>** RF Amplifier from <u>Quantic PMI (Planar Monolithics)</u> Description:24 dBm Solid State Multi-Octave Low Noise Amplifier from 24.5 to 26.5 GHz Configuration: Module with Connector, IC/MMIC/SMT Type: Low Noise Amplifier Sub-Category: SSPA, Multi-Octave Amplifier Frequency: 24.5 to 26.5 GHz Gain: 38 dB Noise Figure: 8 dB Pulsed/CW: CW	

			P1dB: 24 dBm	
12.	Coaxial Cable		**Type:** Coaxial cables can be flexible, semi-rigid, hand formable i.e they hold the space in which they have been bent. **Impedance:** Most cables are 50 Ohms or 75 Ohms. It is important to select the cables based on the impedance of the system or device it is being connected to. **Cutoff Frequency:** This is the maximum frequency up to which the cable can be used. Most cables are designed to operate up to a particular frequency beyond which the losses increase and the cable no longer operates properly. **Dielectric:** This the dielectric material of the cable. The dielectric material used has an impact on the cutoff frequency of the cable and the power level that the cable can support. **Times Microwave Systems - PF130** RF Cable from Times Microwave Systems	RF & Microwave Coaxial Cables are used to transmit high-frequency signals from one point to another. These cables are available in a number of form factors like Flexible, Rigid, Semi-Rigid, Hand Formable and Cryogenic. everything RF has listed RF and Microwave cables from the leading manufacturers and made them searchable by specification.

			Description:52.3 GHz Phase Stable Flexible Cable Type: Flexible Application: Airborn, Ground, Sea, Space Cut off Frequency: 52.3 GHz Propagation Velocity: 83 % Outer Conductor Material: Silver Plated Copper Strip Impedance: 50 Ohms Inner Conductor: Oxygen Free Silver Plated Copper Jacket Material: FEP	
13.	Oscillators		- Frequency 1MHz - 85MHz - Output Waveform HCMOS - Package Type Surface Mount - Dimension 9 x 14 mm - Grade Military	Untuk membangkitkan listrik berfrekuensi tinggi

14.	Modulators		**Type of Modulation:** There are many types of modulation - BPSK, DSB, GMSK, I/Q, I/Q Vector, QAM, QPSK and SSB. **RF Frequency:** The frequency of the RF Signal that is used to modulate the input signal. **Power:** The power of the RF Signal that will be sent out of the modulator. everything RF has created the largest database of RF Modulator products. Complete product catalogs from the leading manufacturers were standardized and complited in to one database which is searchable by specification. **Narda-MITEQ - SSM0812LC2MDA** Modulator from Narda-MITEQ Description:SSB Modulator from 8 to 12 GHz Connectors: SMA, SMA - Female Package Type: Module with Connectors Modulation Type:	An RF Modulator is a device that modulates an input baseband signal using a carrier RF frequency/signal. The modulated signal with the baseband information can now be sent via a radio channel to a receiver, where the signal will be de-modulated and processed further.

			SSB RF Frequency: 8 to 12 GHz Carrier Suppression: 20 to 30 dBc IF Frequency: 20 to 40 MHz LO Frequency: 8 to 12 GHz	
15.	Receivers		• Part Number R5000 • Manufacturer IZT • Description Digital Receiver from 9 kHz and 18 GHz **General Parameters** • Frequency	The IZT R5000 from IZT GmbH is a digital receiver that operates from from 9 kHz and 18 GHz. It supports up to 100 MHz of instantaneous bandwidth. The received signals are available in up to four configurable wideband I/Q Data channels with user configurable sample rates and independent center frequencies. The wideband channels can provide between 5 MHz and 100 MHz of I/Q data. The receiver has 32 additional channels with up to 1 MHz

			9 kHz to 18 GHz	bandwidth each can be used for narrowband monitoring, demodulation or analysis functionality in software.
16.	Circulators		**Selecting RF Circulators** When selecting an RF circulator it is important to know the parameters that you need to specify. Below are a list of the key parameters to look out for when selecting a circulator: **Frequency (MHz):** This is the frequency range over which the device can operate while providing minimal insertion loss and a constant level of isolation. **Insertion Loss (dB):** The signal loss of the incident signal from one port to the next. It is the difference in the incident power at one port to the power received at the next port. **Isolation (dB):** This is a measure of signal levels at the adjacent ports of the circulator i.e when a signal is incident at port one and transmitted to port 2, the isolation is measured between port 1 and port 3. The higher this value the better the performance of the circulator. It is measured in decibel (dB).	An RF Circulator is a 3 Port ferromagnetic device used to regulate the signal flow within a circuit. It transmits an input signal in one direction – A signal that enters port 1 is transmitted to port 2 and isolated from port 3, a signal incident at port 2 is transmitted to port 3 and isolated from port 1, a signal incident at port 3 is transmitted to port 1 and isolated from port 2. Circulators are typically designed to have minimal loss when transmitting an input signal from one port to the next.

			Power (W): This is the level of power that the device can handle while maintaining its electrical characteristics. **Package Type:** These are ferromagnetic devices and are packaged in various form factors – Drop-In, Surface mount, modules with connectors etc. **<u>Renaissance Electronics Corporation - 3B9BC</u>** RF & Microwave Circulator from <u>Renaissance Electronics Corporation</u> Description:100 W Coaxial Circulator from 8 to 14 GHz Type: Coaxial Circulator Application Type: Ku Band, X Band Frequency: 8 to 14 GHz Average Power Forward: 100 W Average Power Reverse: 10 W	

			Isolation: 15 dB Insertion Loss: 0.75 dB VSWR: 1.50:1 Connector: SMA, SMA - Female, SMA - Male	
17.	Wave Guide		• Type Power Amplifier • Configuration Module with Connector, Waveguide • Application Communication • Industry Application Radar • Frequency	Berfungsio untuk menghantarkan gelombang elektromagnetik dan mampu mengarahkannya kea rah tertentu

			34 to 36 GHz • Gain 35 dB • Output Power 37.78 to 38.45 dBm • Output Power 6 to 7 W • Grade Commercial • Saturated Power 37.78 to 38.45 dBm • Saturated Power 6 to 7 W • VSWR 2.0:1 • Sub-Category	

			SSPA • Supply Voltage 8 Vdc • Quiscent Current 4.5 A • Input Connector 2.92 mm • Output Connector WR-28 • Cooling Options Heat Sink and Fan Assembly • Waveguide WR-28 • Note Control Voltage: 0 to 5 Vdc ; Switching Speed:100 ns	

18.	Radar		Type Low Noise Amplifier • Configuration Module with Connector • Industry Application EMC, Military, Radar, Test & Measurement • Frequency to 18 GHz • Gain 35 dB • Gain Flatness ±2.5 dB • Noise Figure 1.3 to 1.5 dB • Grade Military, Commercial • VSWR 2.5:1 • Supply Voltage 11 to 24 V • Current Consumption 350 mA • Transistor Technology pHEMT • Connectors SMA - Female • Operating Temperature -40 to 70 Degree C • Storage Temperature	Berfungsi sebagai Penguat Kebisingan Rendah yang beroperasi dari 1 hingga 18 GHz.

19.	Detectors		• Type Zero-Bias Schottky Diode • Polarity Negative • Frequency 8 to 12 GHz • Max Input Power 17 dBm • TSS -50 dBm • Frequency Flatness ±0.5 dB • Voltage Sensitivity 800 mV/mW • Video Bandwidth 2 MHz	An RF detector monitors or samples the output of an RF system to develop a dc output voltage proportional to the power at that point. They are used to measure and control RF power in wireless systems. RF Detectors are critical for most commercial transmitters where the output power level must not exceed certain values based on the Federal Communications Commission regulations. Power detectors are used to provide feedback to the wireless systems based on which they regulate the output power.

			• VSWR 2.50:1 • Impedance 50 Ohms • Package Type Connectorized • Input Connector SMA - Male • Output Connector SMA - Female • Operating Temperature -55 to 100 Degree C • Note Optional Output Polarity : Positive (suffix "P")	